MD RAKIBUL HASAN HIMAL

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Urban planner and GIS/remote sensing specialist with 5+ years of experience at Bangladesh's capital development authority (RAJUK) and in planning consultancy, working on **land-use regulation, land readjustment, transit-oriented development and urban regeneration** in Dhaka. I am drafting regulatory rules for land readjustment and densification around transit nodes, and I lead drone mapping of about 120 sq km to detect unauthorised land development. Trained in data science (M.Sc.) with peer-reviewed publications in transport safety.

Research interests: housing and land policy in Global South cities; density and floor area ratio (FAR) regulation; development rights, land readjustment and land value capture; informal housing and tenure; transit-oriented development and housing–transport links; GeoAI, UAV photogrammetry and remote sensing for monitoring land development.

Academic Qualification

Master of Science in Data Science and Analytics
East West University, Dhaka
CGPA: 3.80 out of 4.00, passing year: February, 2026

Bachelor of Urban and Regional Planning (BURP)
Bangladesh University of Engineering and Technology (BUET), Dhaka-1000
CGPA: **3.55 out of 4.00**, Merit Position: 7th out of 30, Passing Year: 2019

Higher Secondary Certificate (HSC), Notre Dame College, Dhaka
Passing Year: 2014, GPA: 5.00 out of 5.00

Research Experience and Projects

Undergraduate thesis, BUET, 2019: "A Study on the Problems of Bus Terminals in Dhaka City: Functional Arrangement and Managerial Issues," supervised by Professor Md. Musleh Uddin Hasan, Department of Urban and Regional Planning, BUET.

Graduate capstone (7 credits), East West University, 2025: "Exploring Factors Influencing Fatal Traffic Crashes in the United States (2015–2022): A Spatial and Non-Spatial Clustering Approach," supervised by Dr. Md. Rezaul Karim, Adjunct Professor, EWU.

Deep learning course project, East West University: "A Privacy-Preserving Federated Ensemble Learning Approach for Malaria Detection."

Research collaborator, Transport and Pedestrian Safety Research Group, Department of Urban and Regional Planning, BUET (led by Niaz Mahmud Zafri), 2019: contributed to research on pedestrian crossing behaviour and driver yielding at Dhaka intersections, resulting in two co-authored articles in *Accident Analysis & Prevention* (2020) and *Journal of Safety Research* (2022).

List of Published Journal Articles

Zafri, N. M., Tabassum, T., **Himal, M. R. H**., & Sultana, R., & Debnath, A. K. (2022). Effect of pedestrian characteristics and their road crossing behaviors on driver yielding behavior at controlled intersections. *Journal of Safety Research*, **Elsevier**. [**DOI**]

Zafri, N. M., Sultana, R., **Himal, M. R. H**., & Tabassum, T. (2020). Factors influencing pedestrians' decision to cross the road by risky rolling gap crossing strategy at intersections in Dhaka, Bangladesh. *Accident Analysis & Prevention,* **Elsevier**. [DOI]

Conference Proceedings and Presentations

Himal, M. R. H, Zaman. M. Y. Hasan, M. M. U. (2019). A study on the problems of bus terminals in Dhaka city: functional arrangement and managerial issue. Article published in the proceedings of *International Conference on Urban and Regional Planning (ICURP 2019)*, Bangladesh Institute of Planners (BIP), Dhaka, Bangladesh. October 5 - 6, 2019.

Professional Experience

Assistant Town Planner, Plan Preparation Section,
RAJUK (Capital Development Authority of Bangladesh)

May 2023 to Present

- Currently engaged in drafting the **"Right Conversion Scheme Rules"**, a regulatory framework guiding **land readjustment, mixed-use densification, and urban redevelopment** around mass transit nodes to foster compact and accessible urban form
- Leading the application of **drone-based survey and photogrammetric data processing** under the GIS Lab initiative, producing high-resolution orthophotos covering about 120 sq km to monitor illegal land development and urban sprawl in Dhaka's peripheral areas, thereby strengthening RAJUK's urban monitoring and compliance systems
- Played a central role to the **"Project for Development of Policy and Guidelines for Transit-Oriented Development (TOD) along Mass Transit Corridors"**, which culminated in the officially **gazetted TOD Guidelines for Dhaka**, integrating transit, land use, and sustainable design strategies
- Serving as **Project Manager** for the *"Establishment of GIS/RS Lab for Smart City Planning System and Strengthening Organizational Capacity"* (in collaboration with **ESRI Bangladesh**), focused on developing a city-scale geospatial planning framework for **data-driven decision-making** in Dhaka.
- Developed project documents for **Dhaka Urban Regeneration project**
- Contributed to the **Climate-Resilient Green Action Plan (CRGAP)** project under the **Dhaka Urban Regeneration Program**, supported by the **City Climate Finance Gap Fund (GIZ and EIB)**. Involved in spatial data analysis, stakeholder mapping, and policy formulation for **climate-adaptive regeneration** in **Hazaribagh and Lalbagh**, promoting green infrastructure, drainage improvement, and heat island mitigation
- Train colleagues, including engineering and architecture staff, in **advanced GIS, drone data processing and archive digitisation**

Urban Planner, Sheltech Consultants (Pvt.) Ltd *February 2020 to April 2023*

- Supported the **Bangabandhu Sheikh Mujib Shilpa Nagar Master Plan** (World Bank–BEZA), using **remote sensing and geostatistical tools** to assess land use dynamics and infrastructure expansion
- Performed **geospatial and land cover modeling**, **risk-sensitive land use planning**, and **environmental impact assessments (IEE)**
- Contributed to multiple **feasibility studies and master plans** for pourashavas, universities, IT parks, and economic zones across Bangladesh (LGED, BEZA, UDD, Hi-Tech Park Authority)
- Developed **comprehensive GIS databases**, conducted **spatial accessibility and network flow analyses**, and optimized transportation routes for urban growth management
- Prepared **Environmental Management** and **Climate Change Adaptation Plans** integrating resilience-based planning principles

Intern, Dhaka South City Corporation (DSCC). *September 2018 to October 2018*

- Prepared landscape design and reports for "Golap Shah Majar Intersection Road, Dhaka" after conducting survey

Training, Honors and Awards

- **Training on Building Climate Resilience for Bangladesh Cities and Towns** – 20 days (May 2024), *Erasmus University Rotterdam*, held in **Dhaka**; focused on nature-based solutions for climate mitigation.
- **Urban Development Methods: Land Readjustment and Transit-Oriented Development (TOD)** – 36 days (Nov–Dec 2024), *Japan International Cooperation Agency (JICA)*, held in **Japan**; focused on urban redevelopment strategies for rapidly growing cities.
- **Drone Operation and Application for Urban Monitoring** – 3 days, *RAJUK Capacity Development Program*.
- **Advanced LiDAR Technology for Agricultural Transformation and Sustainable Development** – 3 days, *Specialized Technical Training Program*.
- **International Recognition:**
  - *"Climate-Resilient Urban Regeneration in Dhaka"* selected by **Local Governments for Sustainability (ICLEI)** under the *Global Platform Short Stories on Climate Action* initiative (2025).
  - **RAJUK GIS & RS Lab Web GIS Platform** (as Project Manager) received the **SDG Breakthrough Innovation Award** at the **4th SDG City Awards**, CityNet (2025), for the case study *"Digital Planning for Inclusive Growth: A Scalable Web GIS Platform Shaping Dhaka's SDG-Driven Urban Future"*; also selected among CityNet's *Top 100 Best Practices from 50 Asia-Pacific Cities*.
- **Medha Lalon Scholarship, East West University (2023),** Awarded for outstanding academic performance.
- **Dean's List Scholarship, BUET (2016**) – Awarded for outstanding academic performance.
- **Board Scholarship (General Grade)** – Secondary School Certificate (SSC), 2012.

Technical Skills

Spatial Analytics and Urban Modeling	ArcGIS Pro/ Enterprise, QGIS, GEE
Data Science & Programming	Python (GeoPandas, scikit-learn, PyTorch), R, SQL, Javascript (GEE API)
Design & Simulation:	AutoCAD, SketchUp, CityEngine, Adobe Illustrator, Lumion
Quantitative & Qualitative Methods	Statistical modeling (SPSS, Excel), Survey design, Participatory mapping
Visualization & Reporting	Microsoft Office, Adobe Creative Suite, Power BI, Tableau

Linguistic Proficiency

English (Full Proficiency), **Bangla** (Native Language); **IELTS**: Overall 7.5 (Listening 9.0, Reading 7.5, Speaking 7.0, Writing 6.5)

Reference

1. **Niaz Mahmud Zafri,** Assistant Professor, Department of Urban and Regional Planning, Bangladesh University of Engineering and Technology (BUET), Dhaka. Email: **zafri@urp.buet.ac.bd**
2. **Mahfuja Aktar**, Town Planner, Plan Preparation Section at Capital Development Authority (RAJUK), email: **shahanabuet98@gmail.com**